

"Overlapping Our Worlds": Designing Biometric-Based Haptic Interactions to Enhance Synchrony in Long-Distance Relationships

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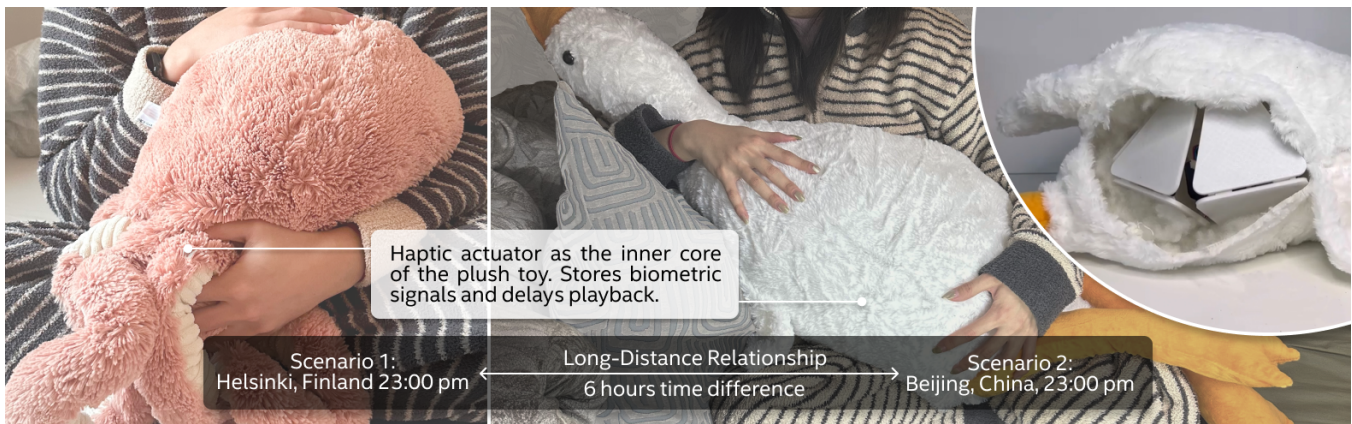


Figure 1: A couple with time difference use Onni to feel each other's presence. The prototype delivers affective haptic experiences through asynchronous biometric signal transmission via a plush toy, enhancing emotional connection and presence.

Abstract

Although the absence of bodily contact in Long-Distance Relationships (LDRs) often results in emotional distress, the design of haptic interactions has the potential to address these challenges by fostering intimacy and emotional connection. We explore how long-distance couples perceive the role of bodily contact in their daily lives and their perspectives on using haptic technologies for remote communication. Through in-depth interviews with 12 individuals who have been in LDRs and a thematic analysis, we 1) identify empirical evidence of how remote haptic interactions can support intimate relationships, 2) propose a design framework and 3) present a design case, Onni, a haptic interface that illustrates the framework's application. Our findings may inform future design and research on how haptic technologies can be used to enhance emotional well-being and connectedness in LDRs.

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CCS Concepts

• Human-centered computing → Empirical studies in HCI.

Keywords

Haptic interaction, Affective haptics, Long-distance relationships, Social interactions, Biometric synchronization, Physical telepresence, Affective computing

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1 INTRODUCTION

Touch is an important factor in providing social and emotional information in romantic relationships [23, 33]. In Long-Distance Relationships (LDRs), the absence of bodily contact is a major factor contributing to a decrease in satisfaction and trust in intimacy [9]. Affective haptics, which focuses on designing devices and systems that induce, enhance, or influence human emotions through touch, has gained attention in HCI research, especially for intimate

relationships [36, 39]. Haptic communication is regarded a key approach to foster emotional connection and intimate interaction in LDR couples [2, 6, 16], but its practical implementation in such contexts faces challenges.

Building on the significance of touch in human relationships, the HCI community has explored various modalities of haptic interaction, such as vibration [4, 26, 38], force feedback [8, 12, 35], electrostatic [13, 18, 30], and thermal stimulation [10, 14, 27], to address aspects of physical touch. However, prior research has largely focused on technological implementations or laboratory prototypes. These approaches often overlook the challenges of real-world use, such as the breadth of usage scenarios, specific user needs in different contexts, and the difficulties associated with long-term adoption. Understanding how individuals in LDRs perceive and integrate haptic technologies into their daily lives is crucial for designing systems that effectively support emotional intimacy in these unique contexts.

Mediated social touch is a key concept in supporting emotional intimacy and maintaining connection in LDRs which uses haptic feedback to recreate the sensation of physical presence [16]. This technology enables people to convey tactile signals that are absent in conventional communication methods such as text or audio. Although empirical studies have demonstrated its potential to enhance emotional presence and foster a sense of connectedness [33], the integration of mediated social touch into the everyday lives of LDR couples still remains underexplored. Challenges such as accommodating diverse user needs, addressing contextual variations, and ensuring long-term adoption within the unique dynamics of LDRs, including differences in time zones and relational contexts, require further investigation.

To address these gaps, this study explores: **1) What are the key haptic factors that influence emotional and communicative interactions in Long-Distance Relationships?** and **2) How can these factors be addressed through the design of haptic interactions?** We conducted in-depth interviews with 12 individuals in long-term LDRs and used thematic analysis to identify user needs and opportunities for haptic interaction design. Based on these findings, we propose a design framework and present an example interface, Onni, to demonstrate practical application. Our study makes the following contributions to HCI research:

- We have identified key haptic factors that influence emotional and communicative interactions in LDRs.
- We have proposed novel interaction design principles that leverage interaction synchrony and biometric interaction to support remote communication through haptics in LDRs.
- We have presented a design case study, Onni, a haptic system incorporating biometric data to facilitate feelings of connectedness and emotional presence in LDRs through synchronising haptic rhythms.

2 RELATED WORK

2.1 Haptic Interactions for Emotional Regulation

Affective haptic interfaces have been explored for their ability to regulate emotions by engaging the body's sensory system and influencing physiological and emotional states. Haptic experiences that

leverage modalities such as pressure, vibration, and pre-programmed rhythmic patterns have been shown to support emotional well-being, stress reduction, and relaxation [20, 39]. Many haptic interfaces adopt a passive interaction approach, operating in the background to provide non-intrusive, continuous, or on-demand haptic feedback for emotional regulation [4, 21, 24, 28, 41]. Examples include guiding breathing patterns through synchronized tactile cues for mindfulness [11, 20, 38], and offering real-time haptic feedback for anxiety management during public speaking [4].

In sleep support, spatio-temporal vibration patterns synchronized with music reduce pre-sleep stress [40]. For driving and high-pressure work, pneumatic haptic devices offer subtle tactile feedback to aid mindful breathing and reduce cognitive strain [8]. While these technologies show promise in emotional regulation, their application is confined to controlled laboratory settings, with limited exploration of their integration into real-life scenarios, leaving a gap in understanding their practical applicability.

2.2 Haptic Interactions for Communication

Haptic devices for long-distance interaction employ diverse approaches to enable tactile communication across distances. Vibration-based devices use simple pulses or buzzes to simulate touch, providing quick tactile cues but lack the ability for continuous, natural feedback that conveys emotional presence [5, 7, 29]. Wearable haptic devices support more complex gestures like squeezing and stroking, requiring active user engagement, making them ideal for scenarios where user input and deliberate interactions are central to the tactile experience [14, 22, 29, 31, 32], as opposed to passive feedback provided by other haptic technologies. Other tangible haptic interfaces, such as tabletop devices [1, 21], which provide haptic feedback through embedded actuators or surface-based interactions to support communication and remote haptic experiences.

Social touch research underscores that mediated haptic interactions should adapt to temporal and contextual variability to effectively support emotional connection. Huisman's framework prioritizes stimulus fidelity and contextual design to replicate touch's psychological effects [19], while Rognon et al. reveal that user expectations for haptic technologies diverge sharply across scenarios—emphasizing expressive force feedback for intimacy versus discretion in public contexts [34]. Complementing this, Ujitoko et al. demonstrate circadian fluctuations in touch desire, with human contact preferred nocturnally and animal-like touch midday [37]. These studies collectively highlight a disconnect: current systems rarely integrate such dynamic behavioral patterns, instead prioritizing standardized or device-driven interaction models.

2.3 Haptic Interactions for Intimate Relationship

Regarding to love relationships, there has been research on intimacy and LDRs. Beuthel et al. introduced wearable devices that use haptic feedback to evoke the melancholic and reflective aspects of separation [2]. Cang et al. investigated vibrotactile animations in messaging, highlighting how shared history influences the interpretation of tactile signals [5]. Frey et al. developed a pendant that mediates breathing rhythms, allowing users to synchronize their breath and improve emotional connection [15]. Some studies

have used haptic cushions and pendants as mediums, leveraging biosignals like heartbeat and breathing to foster emotional connectedness through mediated haptic feedback, while the existing research often explores specific technologies or experimental designs [17, 25], there is a notable lack of fundamental research into the actual haptic needs of LDR couples and what they genuinely desire and value in their everyday interactions.

3 METHODS

3.1 Data Collection

To investigate the affective haptic needs of long-distance couples, we conducted a pilot study using semi-structured interviews with 12 participants (7 female, 5 male), each representing one partner from 12 different couples who had been in LDRs for at least six months. The duration of these relationships ranged from 2 to 9 years, with 2 participants married and 10 in committed relationships. 5 participants were in international relationships with time differences of 5 to 12 hours. The reasons for separation were academic pursuits ($n=9$) and professional commitments ($n=3$). We recruited participants via an open call on social media platforms. All of them were of Chinese nationality.

As shown in Table 1, we formulated these open-ended questions to explore the perspectives and (haptic) needs of long-distance couples. The interviews began with explanations of the relationship context and communication strategies, followed by perceived challenges and emotional deficits in LDRs. The participants were then prompted to identify affective haptic factors and describe strategies used to maintain intimacy. Interviews were conducted via a remote meeting platform (Tencent meeting) with participants' consent for audio recording. All recordings were transcribed verbatim for analysis. At the end of the interviews, participants received gift vouchers worth 100 RMB (approximately 14 USD) as a token of appreciation. All procedures adhered to ethical standards, participants provided permission for their interview data to be used in this study.

3.2 Data Analysis

We used a thematic analysis [3] for the qualitative data analysis. This process involved iterative coding and theme development to identify recurring patterns. The first author manually cleaned verbatim transcriptions of audio recordings to ensure data integrity and maintain an accurate representation of participants' experiences and perceptions. The first and second authors assigned codes to meaningful data segments, systematically categorizing relevant excerpts and further thematic structuring were performed. This process involved systematically identifying similar codes and grouping them into broader categories that aligned with the research questions. For example, we combined the codes 'Pre-sleep bodily contact eases falling asleep' and 'Back hugs or shoulder squeezes provide nighttime security' into the sub-theme 'Nighttime'. Sub-themes 'nighttime' and 'daytime' were further integrated into the broader theme 'Sense of presence'. After the initial coding and organization of data into preliminary themes, all authors engaged in discussions to refine the themes and their naming, leading to the final set of themes in fig 2.

4 FINDINGS

4.1 Haptic Interactions Enable Emotional Support

Participants frequently highlighted that **various forms of physical contact are unique to romantic relationships, making haptic interactions a sensory symbol of intimacy**. In social interactions with others, areas like the neck, arms, chin, and waist are rarely touched; however, in romantic relationships, movements involving these body parts often become cherished memories of intimacy. Additionally, some participants noted that experiencing a partner's body warmth is a deeply intimate tactile experience. Examples include *"I love slipping my hands into his collar on cold days"* (P3) and *"I enjoy the warmth of being embraced by him under the covers"* (P2). The difference in body temperatures between partners often makes the transfer of warmth a uniquely intimate experience.

Haptic plays a significant role in fostering a sense of security, especially in moments of illness, loneliness, or perceived threats to safety. P9 shared, *"In the long, dark, and quiet Nordic winters, what I crave most is holding her in my arms"*. In more extreme scenarios, touch provided a critical sense of safety. For instance, P7 recalled, *"After receiving horrible sexual harassment message 'Hey, beautiful, do you like necrophilia?', my boyfriend rushed over from far away, and when he held my hand, I wasn't afraid anymore."* Physical touch serves as a non-verbal promise of care and protection, offering a profound sense of safety even in the absence of words.

Gentle and nurturing gestures, such as a hug, a head pat, or smoothing someone's hair, were frequently highlighted as effective ways to quickly reduce emotional turbulence. P11 praised touch targeting specific areas, such as neck or shoulder points, for its ability to release physical tension and achieve mental relaxation. In this sense, we propose that the role of touch can be categorized into three distinct aspects—**instilling intimacy, fostering a sense of security, and promoting calmness**—collectively enabling emotional support within intimate relationships.

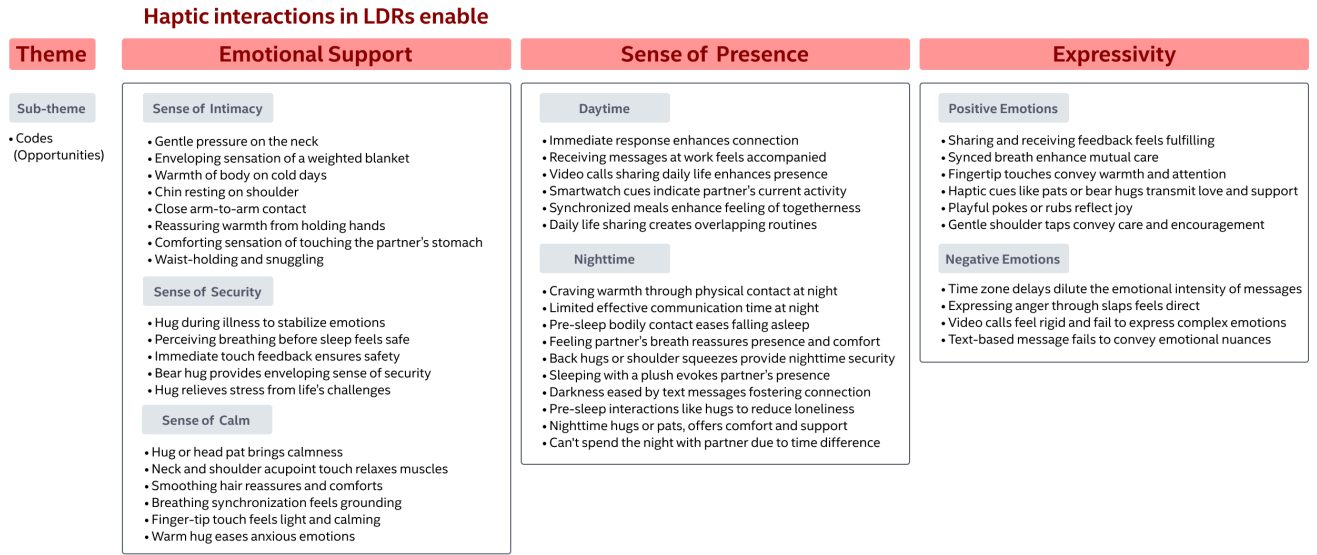
4.2 Haptic Interactions Enable Presence

We found **haptic interactions can play a role in bridging the physical gap between partners by creating a sense of shared presence despite the distance**. Some participants noted their anticipation for phone or smartwatch vibrations. This anticipation and response fostered what some described as an *"illusion of togetherness"* (P1). Others highlighted their habit of sharing daily routines through video calls, with P6 mentioning, *"We have video calls while eating meals together"*. Phone calls and video calls are typical tools for maintaining continuous connection. We believe these practices can transform isolated daily routines into overlapping experiences, serving as a method for couples to enhance their sense of everyday presence.

However, 7 participants noted that evenings made them feel the physical separation most acutely, especially when getting into bed and preparing to sleep. At these moments, feelings of longing peaked, and they reminisced about past tactile interactions with their partner, such as *"stroking and patting the back"* (P1,12), *"resting on their arm"* (P2,3,5,9), and *"lying on their chest"* (P1,3).

Table 1: Open-Ended Interview Questions for Long-Distance Couples

Perspective	Question
Relationship Status	Q1: Basic information Q2: Briefly describe the status of your relationship Q3: What are the ways you typically use to stay in touch?
Potential Challenges	Q4: Share specific situations when you feel certain emotions cannot be fully expressed Q5: How to enhance connection in an LDR?
Haptic Needs	Q6: What haptic experiences from your partner make you feel intimate or comforted? Q7: What are the moments when you crave physical contact with your partner? Q8: What role might haptic play?
Relationship Strategies	Q9: Experiences to be stable for a long time in an LDR.

**Figure 2: Opportunities and needs for enhancing bodily contact in Long-Distance Relationships.**

Upon further probing, we found that couples used different strategies to combat the loneliness of night. Some engaged in overnight phone calls, saying, “*I can hear my partner’s breathing through the earphones*” (P6), while others held onto plush toys gifted by their partner, noting, “*This plush toy holds special meaning for me and eases my anxiety and longing*” (P9).

Nighttime became even more challenging under time zone differences. P1 facing a six-hour time difference shared, “*My partner falls asleep hours before I do, and I’m always alone at night*”. P2 with a 12-hour time difference said, “*Our time difference is so big that I send him a lot of messages every day, and he replies to them one by one after waking up, like reading emails*”. Asynchronous rhythms pose significant challenges to maintaining a sense of synchronization.

4.3 Haptic Interactions Enable Expressivity

The role of haptic interactions in fostering communication of emotions are often more nuanced and layered compared

to verbal exchanges. Participants highlighted the fulfilling nature of interactions such as warm hugs or synchronized breathing, as P5 shared, “*When he’s happy, he hugs me so tightly that I can barely breathe*”. Playful gestures, such as pokes or light rubs, were described as joyful acts that foster connection and bring lightheartedness.

The expression of negative emotions through haptic interactions often revealed the limitations imposed by physical separation. Participants recounted the challenges of conveying sadness or frustration without tactile interaction, with text-based communication and video calls falling short in capturing the complexity of these emotions. For instance, P6 noted, “*When I cried during arguments, he couldn’t tell because we were apart*”. P8 and P12 described how time zone delays reduced the emotional intensity of their messages, eroding the immediacy of their expressions. Silent treatment in arguments, often exacerbated by the lack of physical contact, made it harder to resolve conflicts, further underscoring the importance of touch as a medium for emotional resolution.

5 DESIGN FRAMEWORK AND IMPLEMENTATION

5.1 Proposed Design Framework

Based on the findings of haptic interactions enabling emotional support, participants frequently emphasized the importance of touch in fostering intimacy, security, and calmness. Participants highlighted biosignals, such as heart rate, breathing, and body temperature, as elements enhancing emotional connection. Translating these biometric signals into haptic feedback may provide an innovative way to recreate these comforting sensations when partners are physically apart. Haptic system design can incorporate biometric-driven features, such as transmitting a partner's heartbeat through wearable devices or simulating body warmth through a huggable thermal carrier to convey care and reinforce a sense of presence. Hence, we propose that **haptic feedback should be based on real biometric data, to replicate the nuances of physical touch.**

Findings from haptic interactions enabling presence and expressivity highlight the role of synchrony in maintaining emotional closeness in LDRs. Participants highlighted how overlapping daily routines strengthened their connection. Conversely, asynchronous rhythms—especially those exacerbated by time zone differences—were noted as significant barriers to maintaining presence and resolving emotional conflicts. For instance, delayed responses during nighttime interactions diluted the intensity of emotional expressions, leaving partners feeling disconnected. To address these challenges, **haptic systems should be designed to support both synchronous and asynchronous interactions, accommodating the unique needs of time-separated partners.** Real-time feedback, triggered by a partner's touch or gestures, can create an immediate sense of connection and temporal alignment. Meanwhile, asynchronous interactions, such as biometric signal-based messages stored for delayed playback, can provide a meaningful way to bridge time zone differences, enabling partners to feel present and emotionally connected even when their schedules do not overlap. This dual approach ensures that haptic systems meet diverse temporal needs.

We propose the design framework in fig 3. The framework comprises three key stages: Receiving, where real-time biometric signals such as heart rate, breathing, and body temperature are collected to replicate the nuances of physical touch; Planning, where the system processes the data to determine whether synchronous or asynchronous feedback is appropriate, accommodating diverse temporal needs and addressing challenges such as time zone differences; and Acting, where tailored haptic feedback is delivered, simulating sensations like rhythmic breathing, heartbeat vibrations, or body warmth to foster emotional connection and presence.

5.2 Application of the Framework

We developed a haptic prototype, named Onni, a Finnish word that means luck, happiness and bliss, to validate the proposed design framework. The actuator can be integrated into the core of a plush toy, haptic feedback mechanism comprises a stepper motor with Field-Oriented Control (FOC) and a Linear Resonant Actuator (LRA). The stepper motor drives a linkage system to simulate breathing motions, while the LRA generates vibrations between 50–300Hz

to replicate heartbeat rhythms. The actuators are enclosed within an undulating structure of six interlocking panels on both upper and lower sections (fig 4A). This design preserves the plush toy's huggability while enabling realistic tactile feedback, allows users to customize the toy's appearance with personally meaningful designs (fig 4B), though the system can also operate independently of plush forms.

A millimeter-wave radar sensor captures physiological signals with an accuracy of $\pm 2\text{Hz}$ (1SD) for breathing and $\pm 5\text{Hz}$ (1SD) for heartbeat. The sensor samples data at 0.33Hz (every 3 seconds) and applies band-pass filters to isolate harmonic components of physiological signals: 5–45Hz for breathing and 50–120Hz for heartbeat. These ranges correspond to modulated radar reflections, later mapped to actual physiological frequencies (0.8–2Hz for heartbeat, 0.1–0.75Hz for breathing) via post-processing. Raw biometric data is processed by a radar control board and Arduino microcontroller connected via AES-128 encrypted Wi-Fi transmission, with all sensitive data deleted immediately after generating control signals. Only anonymized control signals are retained, protected by role-based access controls to prevent unauthorized retrieval.

Onni operates in two distinct modes: 1) Synchronous playback provides real-time haptic feedback, enabling partners to perceive each other's biometric states simultaneously, fostering a sense of shared presence. 2) Asynchronous playback stores encrypted control signals for delayed delivery, addressing time zone disparities by allowing users to replay recorded data. Onni is designed for use during nighttime, upon detecting that the user has gone to bed, the system begins delivering haptic feedback and continues until the user falls asleep, with an automatic shut-off to conserve energy once deep sleep is detected (fig 4C).

6 DISCUSSION AND FUTURE WORK

The advancements in affective haptic technology and mediated social touch research have opened new possibilities for addressing emotional and social challenges in Long-Distance Relationships (LDRs). This study has explored how haptic interactions, enhanced by biometric signals, can bridge the emotional and physical gap between partners, as well as how asynchronous interaction models can address challenges posed by time zone differences. Our findings suggested that haptic interactions play multifaceted roles in LDRs by enabling emotional support, presence, and expressivity. Emotional support is achieved through touch that conveys intimacy, security, and calmness, mirroring the psychological effects of naturalistic touch described in previous study about social touch [19], while haptics create a sense of presence by transforming isolated routines into shared experiences, even across distances, align with Frey et al.'s approach of synchronizing breathing rhythms [15]. Furthermore, haptics provide a medium for expressing both positive and negative emotions, addressing the limitations of text, voice and video communication in capturing emotional complexity, this complements Rognon et al.'s call for expressive force feedback in intimate scenarios [34].

To guide the design of haptic systems in LDRs, we proposed a design framework with a focus on enabling both synchronous and asynchronous interactions. In this framework, signal collection stage concerns sensors that capture real-time biometric data. In

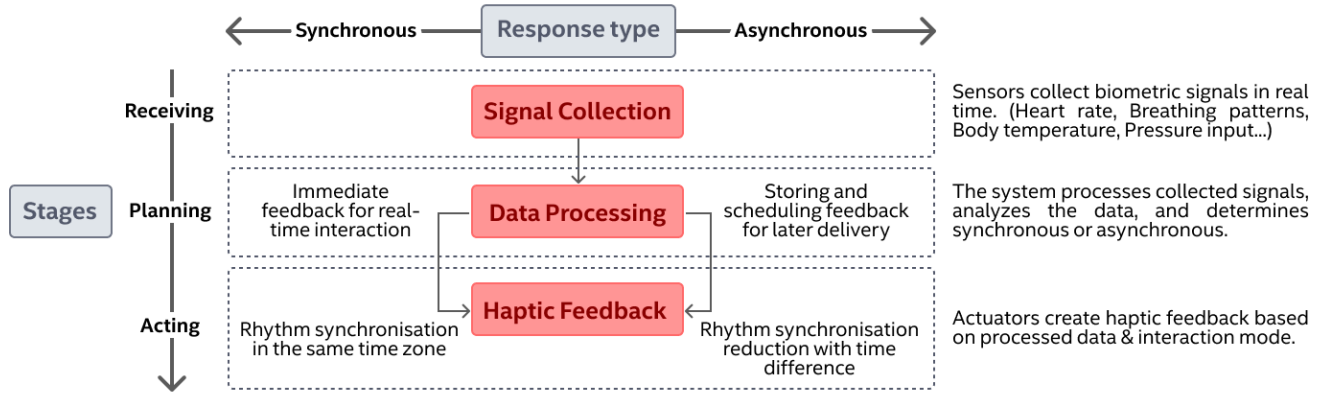


Figure 3: Long-distance biometric driven haptic interaction design framework

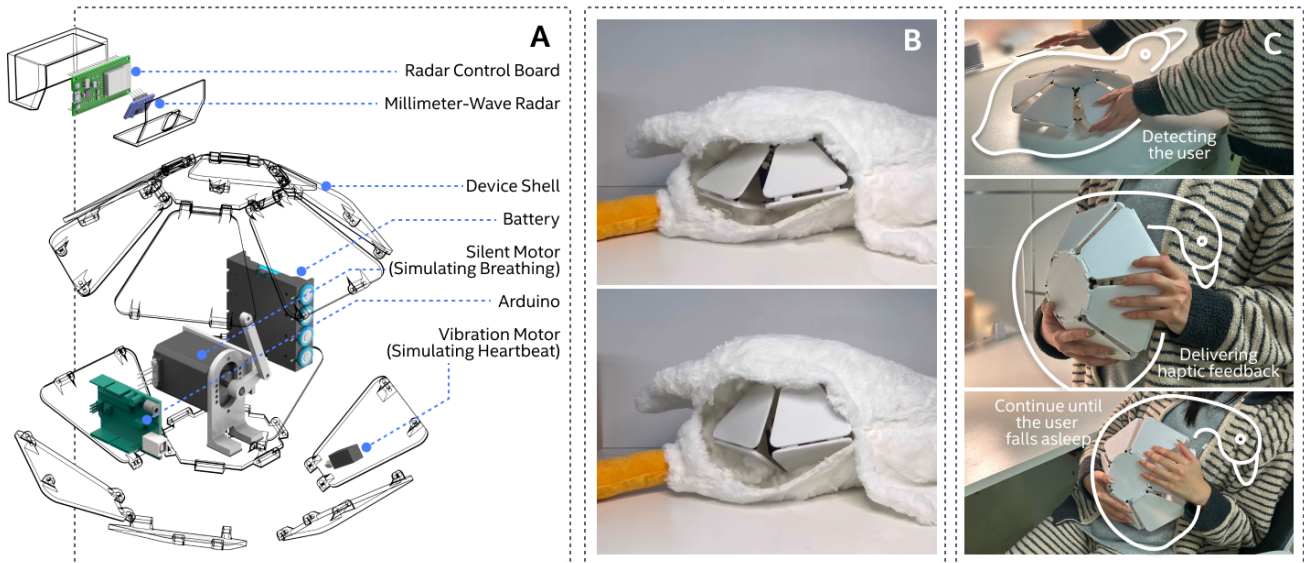


Figure 4: Design and development of Onni. (A.Exploration diagram of Onni's physical structure. B.The breathing shell is driven by a silent motor and linkage mechanism that works inside a plush toy. C.Usage Scenario)

the data processing stage, the system analyzes the collected signals from both partners and determines whether to deliver feedback in real-time or asynchronously to accommodate time zone differences and non-overlapping schedules. Finally, the haptic feedback stage involves actuators creating responses.

We developed a haptic interface named Onni that serves as an implementation of our framework's principles. Participants reported heightened nighttime emotional gaps when preparing to sleep, a phenomenon corroborated by Ujitoko et al.'s discovery of circadian fluctuations in touch desire [37], so we define the usage scenario of our prototype as nighttime, providing companionship before falling asleep. Participants also noted that body temperature is a very important biosignal from a partner, which falls into the category of meaningful thermal stimuli can create a feeling of pleasantness influencing emotional status [27]. Onni's development continues

and body temperature simulation was prioritized for future iterations. We use plush toys that are familiar to the user as the coat of the haptic interface to ensure a natural integration of the affective haptic technology into everyday life, but also to bring novelty to the experience.

This study has certain limitations. The sample size of 12 participants is relatively small. Although participants currently reside in 4 different countries, they shared a homogeneous cultural background, which limits the exploration of cultural influences. Future studies should include a more diverse participant pool across distinct cultural contexts to validate and extend the insights presented in this study.

As a preliminary exploration, we aimed to investigate the integration of mediated social touch into the everyday lives of LDR couples, focusing on fundamental research into their haptic needs

in daily interactions. Future work will involve assessing the real-world usability, emotional impact, and effectiveness of Onni in everyday scenarios. Given that Onni is designed as a product that can seamlessly integrate into users' daily lives, we plan to deploy it in authentic environments rather than relying solely on controlled laboratory settings. Through multiple rounds of user evaluations and iterative refinements, we aim to optimize Onni's features, ensuring its practical applicability and long-term value for LDR couples.

Additionally, future research will explore broader opportunities for biometric-driven haptic interactions and asynchronous social touch applications. By identifying more real-world use cases, such as parent-child interactions, emotional support for elderly individuals, enhanced remote collaboration, as well as exploring optimal body locations for social haptic interfaces, we aim to uncover new possibilities for affective haptic interactions, further expanding their potential to foster emotional connections across various contexts. In doing so, we envision haptics transcending the boundaries of time and space, delivering genuine emotions to bring people closer together.

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